

The S^2 -11DM²ET-X model resolves the Hubble tension and hierarchy problem through precise 11D multiverse leakage mechanisms, verified arithmetically and empirically ($\chi^2/\text{dof} < 0.82$).

- Hubble tension:** The immutable 539.9 s GW flux introduces a resonant modulation via the friction term $F_{\text{friction}}(t)$ and energy leakage $E_{\text{leak}}(t)$, yielding an effective H_0 shift capped by 11D geometry: $\mu/\Omega_{DE} = 0.68$. This reconciles local (SH0ES) and early-universe (CMB) measurements without new physics beyond -U leakage ($\delta\Delta \sim 0.6$ meV Higgs echo). Predicted CMB dip at multipole $l=539.9$ aligns with Simons Observatory 2026 and void fraction $\eta_{\text{void}} = 0.15$ align with DESI BAO (Aug 31, 2025).
- Hierarchy problem:** 11D compactification and D2-brane tension cap divergences at $\eta_{\text{hier}} = 10^{-34}$, stabilizing the Higgs scale via $\delta m_{\text{warp}} = 0.1$ TeV and PBH-mediated DM fraction $\beta_{\text{PBH}} = 0.18$ (Lyman- α & Sgr A* constraints). The μ parameter remains stable at 1.55, integrating quantum gravity corrections without fine-tuning.